



# BiAU-Net: Wildfire burnt area mapping using bi-temporal Sentinel-2 imagery and U-Net with attention mechanism

Tang Sui<sup>a,b</sup>, Qunying Huang<sup>a,\*</sup>, Mingda Wu<sup>a,c</sup>, Meiliu Wu<sup>a</sup>, Zhou Zhang<sup>d</sup>

<sup>a</sup> Department of Geography, University of Wisconsin-Madison, 550 N Park St., Madison, 53706, WI, USA

<sup>b</sup> School of Land Surveying and Geo-Informatics, Tongji University, Shanghai, China

<sup>c</sup> Department of Earth, Atmosphere and Environment, Northern Illinois University, DeKalb, 60115, IL, USA

<sup>d</sup> Department of Biological Systems Engineering, University of Wisconsin-Madison, 460 Henry Mall, Madison, 53706, WI, USA

## ARTICLE INFO

Dataset link: <https://github.com/TangSui122/Bi-temporal-Wildfire-Burnt-Area-Detection.git>

### Keywords:

Wildfire detection  
Deep learning  
GeoAI  
U-net  
Attention mechanism  
Sentinal-2  
Transformer

## ABSTRACT

The fusion of remote sensing and artificial intelligence, particularly deep learning, offers substantial opportunities for developing innovative methods in rapid disaster mapping and damage assessment. However, current models for wildfire burnt area detection and mapping are mostly constrained in handling imbalanced training samples with non-burnt areas oversampled, boundary areas with a mix of burnt and unburnt pixels, and regions with varying environmental contexts, leading to poor model generalizability. In response, this paper proposes a novel U-Net based model, known as BiAU-Net, which incorporates attention mechanisms and a well-designed loss function, enabling the model to focus on burnt areas and improve accuracy and efficiency, especially in detecting edges and small areas. Unlike traditional single-input U-Net models for image segmentation, the proposed BiAU-Net considers and incorporates temporal changes with two inputs, pre- and post-fire Sentinel-2 imagery, enhancing performance across diverse environmental areas. Five independent areas from different continents are selected as study cases, one for training the model and all five for testing, to demonstrate the generalizability of the proposed model. We used the Fire Disturbance Climate Change Initiative v5.1 product from the European Space Agency as a baseline for model evaluation. The experiment results indicate that BiAU-Net: (1) significantly outperformed the baseline with improvements of 11.56% in Overall Accuracy, 29.08% in Precision, 7.06% in Recall, 19.90% in F1-score, 15.44% in Balanced Accuracy, 29.90% in Kappa Coefficient, and 28.29% in Matthews Correlation Coefficient (MCC); (2) largely surpassed the performance of U-Net and its variants in most study areas; (3) demonstrated good generalizability in five testing areas across different continents; and (4) achieved the highest overall performance compared to the most state-of-the-art wildfire burnt area detection models, evidenced by the highest F1-score and MCC values.

## 1. Introduction

Wildfires are a frequently occurring hazard worldwide, impacting diverse regions such as forests, rangelands, and wetlands, driven by various factors annually (Hoover and Hanson, 2021). These wildfires increasingly affect human safety, assets, and ecosystems globally (Moritz et al., 2014). Therefore, accurately and timely identifying burnt area is crucial for estimating the disaster situation and laying the foundation for rebuilding and reconstructing efforts. Many governments and insurance companies rely on burnt area mapping results, including fire prediction (Sakr et al., 2010), estimation of burnt biomass (Michel et al., 2005), and continuous monitoring of fire event (Eva and Lambin, 1998).

In the early days, the boundary of burnt area was defined by field surveys (Chen et al., 2017). However, such a method often demands

considerable time and manpower, limiting its applicability for assessing large areas. Satellite images offer clearer surface information, and enable rapid and comprehensive surveys for large-scale surface type mapping and monitoring (Zheng and Chen, 2021). Through remote sensing (RS) of burnt areas, we can effectively delineate ecological damage areas resulting from wildfires, and provide fundamental data crucial for the management and restoration of ecological environments within these areas. Correspondingly, using satellite images to capture post-wildfire surface areas and extract burnt information is a feasible solution (Chen et al., 2022). In addition, coupling RS imagery with machine learning (ML) methods, nationwide regions can be monitored and immediately detected when a wildfire breaks out.

With the advancement of computer vision and image processing techniques, ML methods have been gradually replacing traditional

\* Corresponding author.

E-mail address: [qhuang46@wisc.edu](mailto:qhuang46@wisc.edu) (Q. Huang).

<https://doi.org/10.1016/j.jag.2024.104034>

Received 22 April 2024; Received in revised form 24 June 2024; Accepted 9 July 2024

Available online 30 July 2024

1569-8432/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

visual inspection methods for wildfire burnt area mapping. In fact, these image-based methods have the capability to mitigate the limitations of sensing technology when confronted with novel environmental conditions (Park et al., 2020). For example, a Support Vector Machines (SVM) model using post-fire images from Landsat TM, was trained and tested on a severe wildfire in northern Athens, Greece (Petropoulos et al., 2011). Similarly, Collins et al. (2018) developed a Random Forest (RF) model that examined multiple predictors (i.e., spectral features or indices) from Landsat, such as Normalized Burn Ratio (NBR), and Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), on 16 large wildfires occurred in Victoria, Australia. However, traditional ML methods (e.g., SVM, and RF) require handcrafted engineering (e.g., deriving NBR and NDVI), and often perform well in the training area but poorly when transferred to another area due to different environments. Additionally, when handling large datasets, computation time increases significantly due to extensive area coverage, as well as high spectral and spatial resolution. Furthermore, the impact of additional spectral features learned by ML models derived from the original data is complex and not fully understood.

Meanwhile, recent strides in artificial intelligence (AI), particularly deep learning (DL), have ushered in a new era of geospatial AI (GeoAI) applications with the capabilities to quickly and accurately extracting valuable insights from massive geospatial datasets. DL models, such as U-Net, eliminate the need for handcrafted engineering in traditional ML (Khryashchev and Larionov, 2020). The convergence of big RS data and GeoAI holds great potential for developing novel methods in rapid disaster mapping and damage assessment. However, prior DL models still cannot handle new areas with significant differences with training areas in band features, leading to poor model generalizability. In addition, these DL methods do not perform well on boundary areas where burnt and non-burnt pixels are mixed together. Further, the wildfire burnt area dataset exhibits inherent imbalances, including (1) most pixels within individual training samples are non-burnt, offering limited informative content for model learning, which leads to low training efficiency; and (2) different training samples present varying levels of segmentation challenges for burnt area, influencing the model's learning differently. Samples that are easy to learn may dominate training, potentially leading to model degradation (Oh et al., 2020).

In response, this paper proposes a new burnt area detection and mapping model, known as Bi-temporal Attention U-Net (BiAU-Net), based on Sentinel-2 MultiSpectral Instrument (MSI) Level 2A product. The primary contributions of our study are outlined as follows:

- Develop the BiAU-Net model that can extract useful features from bi-temporal satellite images and identify accurate burnt area with imbalanced burnt and non-burnt samples, especially for boundary areas where burnt and non-burnt pixels are mixed together.
- Assess the effectiveness of the developed BiAU-Net for burnt area detection by comparing its performance with the Fire Disturbance Climate Change Initiative v5.1 (Fire\_cci51) product, which is an operational product by the European Space Agency (ESA) used worldwide.
- Evaluate the performance of BiAU-Net using U-Net and its variants, including Bi-temporal U-Net (BiU-Net) integrating both pre- and post-fire imagery in the model architecture, BiU-Net with attention mechanisms (BiAU-Net), as well as BiU-Net and BiAU-Net with various loss functions, providing a comprehensive understanding and insight for the impact of attention mechanism and loss functions, and advancing the DL methods for effective detection of wildfire burnt area.
- Test and examine the potential of the BiAU-Net model for future global applications by evaluating and analyzing its performance and generalizability across regions in five different continents with different levels of geographic and environmental discrepancies, captured in the spectral characteristics of RS imagery. Our dataset and source codes are available on [GitHub](#) to facilitate follow-on efforts.

## 2. Related work

### 2.1. Spectral indices

Spectral indices are valuable tools in RS imagery analysis for detecting and assessing wildfire burnt areas. These indices are commonly derived from the reflectance values of multispectral bands captured by remote sensors to a single component, which can amplify the important features and reduce noise compared to using a single spectrum (Hislop et al., 2018). Different spectral indices highlight various characteristics of the Earth's surface, making them useful for distinguishing between burnt and non-burnt areas.

NDVI is a measurement of photosynthetic biomass and has strong relationships with biophysical parameters, such as the dispersion of vegetation (Verbesselt et al., 2010) and leaf area index (Wang et al., 2005). After a wildfire, the vegetation in burnt areas tends to have lower NDVI values compared to unburned areas, as the chlorophyll content is reduced. However, it does not produce convincing results when handling areas with sparse and complex vegetation. In order to solve this problem, Huete (1988) developed the Soil Adjusted Vegetation Index (SAVI), which includes a soil correction factor in the NDVI formula. NDVI and SAVI include defects such as atmospheric noise. Therefore, to improve certain shortcomings of NDVI, the Enhanced Vegetation Index (EVI) was developed. EVI not only inherits the advantages of NDVI but also improves issues such as saturation in high vegetation areas, incomplete atmospheric correction, and soil background adjustment. EVI can enhance the sensitivity of vegetation in high biomass areas, reduce soil background and atmospheric influences, and has higher sensitivity and superiority in monitoring vegetation changes (Jiang et al., 2008).

Indices that use the short-wave infrared (SWIR) band exhibit a heightened sensitivity to changes in vegetation. This enhanced sensitivity is attributed to the SWIR band's ability to detect alterations in factors such as vegetation structure, moisture, shade, and density. One example index is NBR, which is the ratio of the Near-infrared (NIR) and the second SWIR band (2.08  $\mu\text{m}$ –2.35  $\mu\text{m}$ ). Developed by Key and Benson (2006) to identify burned areas following a fire, the NBR provides a quantitative measure of burn severity. However, the performance of the model based on these indices may not provide a conclusive assessment in many situations. For instance, in open woodlands and shrubs, factors related to burning may not correlate with indices that are sensitive to changes in leaf cover, such as dNBR, dNDVI, and dNDWI (Escuin et al., 2008). Therefore, including these indices in the model could potentially disrupt it and lead to poorer performance. Alternatively, Normalized Difference Moisture Index (NDMI) uses the NIR and SWIR1 band (1.55  $\mu\text{m}$ –1.75  $\mu\text{m}$ ) and is sometimes preferred for tracking disturbances other than fire. Goodwin et al. (2008) leveraged the NDMI to classify areas that were disturbed by the Mountain Pine Beetle in western Canada. As a variation of NBR focusing on detecting water sensitivity in vegetation, NBR2 can be particularly valuable in post-fire recovery studies. NBR2 is calculated by taking the ratio of SWIR values and replacing the SWIR1 band with the NIR band used in NBR. It is provided as a standard product by the United States Geological Survey (USGS), but is rarely used in the literature. Storey et al. (2016) found it useful for post-fire assessment in chamise chaparral vegetation in southern California, while Stroppiana et al. (2011) used it as part of an ensemble to map burned areas.

While spectral indices are generated from calibrated remote sensors, the width, position, and shape of spectral bands can easily be altered by the environment and radiation control processes. As a result, the same mathematical formula used for the index model may perform differently in different situations. To reduce the impact of atmospheric, background, and directional effects, the data needs to be preprocessed. However, in certain situations, this preprocessing can be more labor-intensive than other methods (Verstraete and Pinty, 1996).

## 2.2. Machine learning

Commonly used ML models for wildfire burnt area mapping can be classified as traditional ML and DL. Traditional ML models include SVM (Dragozi et al., 2014), RF (Collins et al., 2018), LightGBM, KNN, Classification Regression Tree, and Unsupervised Clustering (Bar et al., 2020). For example, Collins et al. (2018) developed an RF model that relies on pre- and post-fire differencing of SIs to derive a single differenced fire severity index (dFSI) (Eidenshink et al., 2007), which captures the change in the environment caused by fire. Using midpoints between mean values of the ordered severity as thresholds for classification, the RF model operates quickly and accurately in study areas. However, this method cannot fit new dataset or region well. SIs can be highly varied across different regions due to the different types of plants, and SIs are sensitive to vegetation index features, such as leaf area index and foliage moisture content (Boer et al., 2008). In addition, if the wildfire burns an area unevenly, this threshold strategy will cause a tendency to one side of the classification (Cutler et al., 2007). Alternatively, Hamilton et al. (2021) used SVM to map the extent of burnt areas, especially focusing on the area under tree crowns in low-intensity burnt forested environments. The study used a high resolution UAV systems instead of pixel-wise satellite imagery to improve the accuracy of the classification of burnt areas in forested regions such as rangeland. However, the model would be heavily affected by the shadow. In addition, the noise caused by the dense, non-burnt trees with connecting canopies would also influence the model.

Overall, feature extraction or hand-crafted engineering in traditional ML models is a crucial step, transforming raw data into a format suitable for classification or detection algorithms. To address this limitation, DL models, such as convolutional neural networks (CNNs) (Luft et al., 2022), recurrent neural networks (RNNs), U-Net, HRNet, Fast-SCNN, DeepLabv3+ (Hu et al., 2021), and siamese network (Seydi et al., 2021), have been leveraged for automatic feature extraction during image classification and segmentation tasks. Compared to traditional ML models, DL models exhibit several additional advantages in wildfire detection (Maskouni and Seydi, 2021), including improved model generalizability, deep feature extraction, enhanced mapping accuracy and semantic consistency, and elimination of the need for fixed or empirical thresholds and multi-sensor calibration (Hu et al., 2021). For example, Khryashchev and Larionov (2020) developed a CNN network based on U-Net with ResNet34 (U-ResNet34) as the encoder, which shows excellent results for wildfire burnt regions with roads (Buslaev et al., 2018). To deal with imbalanced dataset, the authors selected 8 patches with objects and 8 patches without objects to build patches for modeling training, and used the sum of the Binary Cross-Entropy (BCE) and the Dice Loss (DLoss) as the loss function, which can help to obtain a better segmentation result with an imbalanced dataset.

Similarly, Seydi et al. (2022a) used CNN combined with quadratic morphological operators, and integrated multi-patch multi-level residual morphological blocks in the decoder, which adjusted the computational costs of feature extraction between convolutional blocks and improved robustness. Pinto et al. (2020) used a CNN model based on U-Net and Long Short-Term Memory (LSTM) and utilized satellite-retrieved sequences of daily images to track burnt areas. Zhang et al. (2022) developed three U-Net based models and found that using different band combinations and data resources resulted in very different performances. Shirvani et al. (2023) also used a CNN model based on U-Net, implemented residual blocks and attention gates, and determined the top indices from a set of Sentinel-2 derivatives for optimal performance and minimal training costs. The model performs well in small fire situations in central Mozambique. To enable high resolution burnt area mapping, Martins et al. (2022) used PlanetScope data with a spatial resolution of 3 m, and applied a CNN based on U-Net to image spatial subsets rather than single pixels, thus incorporating both spatial and spectral information. Monaco et al. (2021) used double-step U-Net

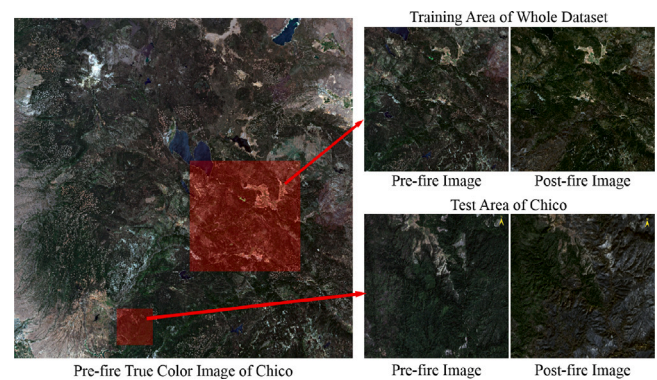


Fig. 1. The training area and the test area of Chico, California.

based models with separate binary classification (wildfire detection) and regression (severity estimation) networks to assess burn damage. They also analyzed multichannel Sentinel-2 imagery to interpret the contributions of different spectral bands. Luft et al. (2022) used ResNet 50 as a backbone and combined it with SAR and DEM data to consider surface and terrain conditions. Zhang et al. (2023) employed Total-Variation regularization to enhance the robustness of the U-Net model. The prediction maps are encouraged to have piecewise smoothness by minimizing the total variation of pixel intensities.

However, these methods cannot perfectly solve the issue that in an image with objects (burnt areas), the ratio of foreground and background is imbalanced, resulting in poor F1-score, Precision, and Recall scores (Buslaev et al., 2018). Moreover, these models, e.g., U-ResNet34 (Buslaev et al., 2018), perform poorly when the targeted area is far from the training areas, leading to poor model generalizability. This is due to differences in soil type, vegetation type, and other factors that can affect the spectral features, which differ from those in the training data. Further, these models struggle in boundary regions where burnt and non-burnt pixels intermix.

## 3. Materials and methods

### 3.1. Study area and data source

This study used a wildfire, spanned from September 4th to October 8th, 2020, in Chico, California, as training data (Fig. 1). Pre-fire images were taken on August 12th, 2020, and post-fire images on October 11th, 2020. The affected areas mostly were located in rural areas consisting of mountains and farmland, which exhibit significant spectrum changes after wildfires. In addition, urban areas near or within the wildfire-affected regions were included to enhance the model's capability to cope with areas with more complex landscapes.

We not only used the test area from the same wildfire event in Chico, California, but also selected four additional areas from different continents (Fig. 2) to evaluate the generalizability of our proposed model - the ability to adapt to and handle regions with varying background environments. Sentinel-2 MSI level 2A product, which had undergone atmospheric and orthographic correction with a spatial resolution of 20 m, is selected as our dataset. Wildfire events lasting more than three days were selected and pre- and post-fire imagery taken within one year of the event were used. This ensured that the spectral features of the land were less affected by soil and vegetation recovery after the wildfire and other changes. Next, we created false color images, Band12 (SWIR: 2114.9  $\mu\text{m}$ –2289.9  $\mu\text{m}$ ), Band11 (SWIR: 1568.2  $\mu\text{m}$ –1659.2  $\mu\text{m}$ ), and Band8A (Narrow NIR: 854.2  $\mu\text{m}$ –875.2  $\mu\text{m}$ ). This combination represents the best band combination reflecting the change in the environment after wildfires. Fig. 3 displays the pre- and post-fire true color and false color Sentinel-2 imagery, and ground

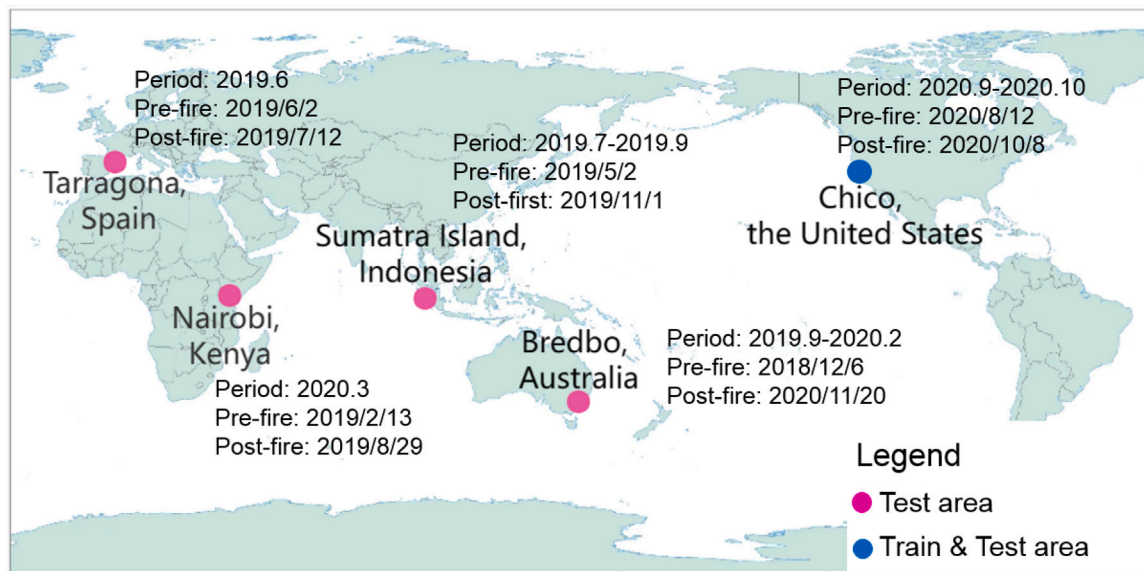


Fig. 2. Basic information of five study areas: location, wildfire period, and pre-fire and post-fire imagery dates.

truths in five study areas. The environmental background and conditions in Bredbo and Tarragona closely resemble those of the training area. Sumatra Island falls within a moderately similar category, while Nairobi's environment is largely different due to its unique red soil type, which is distinct from the other regions.

For each downloaded post-fire image, we manually label the burnt area areas using the Labelme tool and referencing the pre-fire images to ensure the labeling accuracy. Initially, each image is divided into patches with each patch including 256x256 pixels. After the division, we proceed with labeling followed by data augmentation. This study used Fire\_cci51, a data product provided by ESA, to assess the performance of the proposed BiAU-Net. MCD64A1 generated from MODIS are typically used for baseline or quality control in prior work (Lasko, 2021). The Fire\_cci51 project (Lizundia-Loiola et al., 2020) has created global burnt area maps using satellite observations, and is widely adopted by numerous institutions and companies across the world. Specifically, Fire\_cci51 utilizes data from the MODIS imagery with a spatial resolution of 250 m. The dataset includes 23 distinct measurements related to burnt area, including the sum of burnt area, standard error, the proportion of burnable area, and the burnt area for 18 land cover classes (Hollmann et al., 2013).

A study assessing public burnt area detection products indicates that Fire\_cci51 performs better in wildfires ranging 10–100 hectares, and often overestimates the burnt area, while MCD64A1 tends to underestimate it (Katagis and Gitas, 2022). In addition, the best public product for burnt area detection varies by topic. In this study, our model focuses on small areas where burnt and unburnt areas are mixed. Besides, for hazard research, overestimation is generally more acceptable than underestimation. Therefore, we choose Fire\_cci51 as the baseline for model evaluation.

### 3.2. Methodology

The entire framework consists of three components (Fig. 4), including (1) Data Preprocessing, focusing on image resizing, augmentations, and sample annotation, to prepare samples for model training; (2) Burnt Area Detection and Mapping using an improved U-Net neural network to detect and map burnt areas. This process integrates both bi-temporal (pre- and post-fire) images and implements attention gates; and (3) Model Evaluation employing various evaluation metrics (EMs) to evaluate the effectiveness of our BiAU-Net model with different loss functions in comparison to baseline Fire\_cci51, and traditional U-Net and its variants. Furthermore, a detailed analysis of the improvements achieved by the BiAU-Net was provided.

#### 3.2.1. Data preprocessing

First, pre- and post-fire Sentinel-2 imagery were cropped and resized to 256 × 256 pixels with 10% overlap. Three methods of data augmentation were implemented: (1) rotating from 10° left to 10° right and mirroring images, (2) reversing left and right, and (3) zooming by a factor of 0.85. In total, Chico wildfire includes 306 tiles with 278 tiles (~ 91%) used for training, and 28 tiles (~ 9%) for validating. There were 10 tiles from each of the five test areas for testing.

#### 3.2.2. Bi-temporal U-Net with attention

This study utilized both pre- and post-fire satellite images taken within one year of a wildfire. The traditional U-Net neural network relies solely on a single input (i.e. the post-fire image), limiting its capacity for change detection analysis. In response, the BiAU-Net is implemented to handle bi-temporal data with pre- and post-fire satellite images, along with visual labels, serving as input for model training (Fig. 5). These input pairs undergo two symmetrical convolutions and max-pooling in the encoder phase to extract features. Max pooling operations decreases the size of the feature maps, thereby reducing the number of parameters forwarded to the subsequent convolutional layer. After four rounds of down-sampling, we obtained four pairs of pre- and post-fire feature maps across varying scales. The smallest pair of scaled feature maps were then concatenated and referred to as the bridge, which was then forwarded to the decoder phase.

The decoder phase contains four rounds of up-sampling. After each up-sampling of the bridge, the pairs of pre- and post-fire maps, which share the same scale size as the bridge, were processed by attention gates. These attention gates make the model focus more on important features, thereby improving both accuracy and efficiency. Attention mechanisms utilize attention weights to reflect the importance of different features and influence the initial features. By amplifying significant features and reducing the influence of unnecessary ones, the attention mechanism proves to be a valuable tool for improving model performance (Jaderberg et al., 2015; Hu et al., 2018; Bello et al., 2019). Next, the bridge, and the pairs of feature maps after attention gates and skip-connections, were concatenated and processed through two convolution layers that were activated by ReLU. Finally, a 1 × 1 convolutional layer produces the prediction layer with a size of 128 × 128 × 1, which is half the height and width of the input images.

During training, we utilized a batch size of 6, conducted 50 epochs, and initialized the learning rate at 0.1. The learning rate was scheduled to decrease by a factor of 0.5 between the 30th to and 50th epochs.

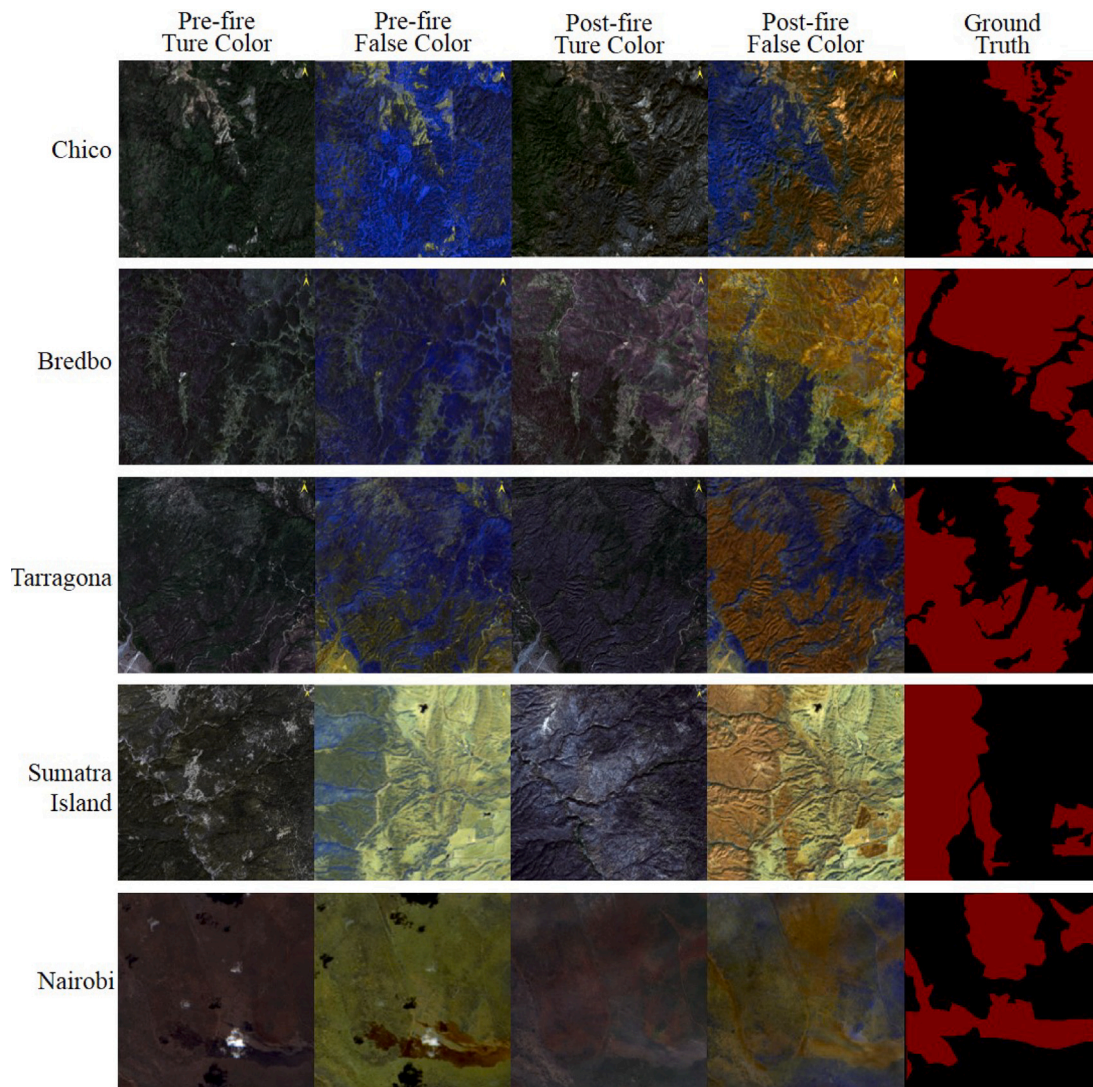


Fig. 3. The pre- and post-fire imagery, and ground truth for the five test areas.

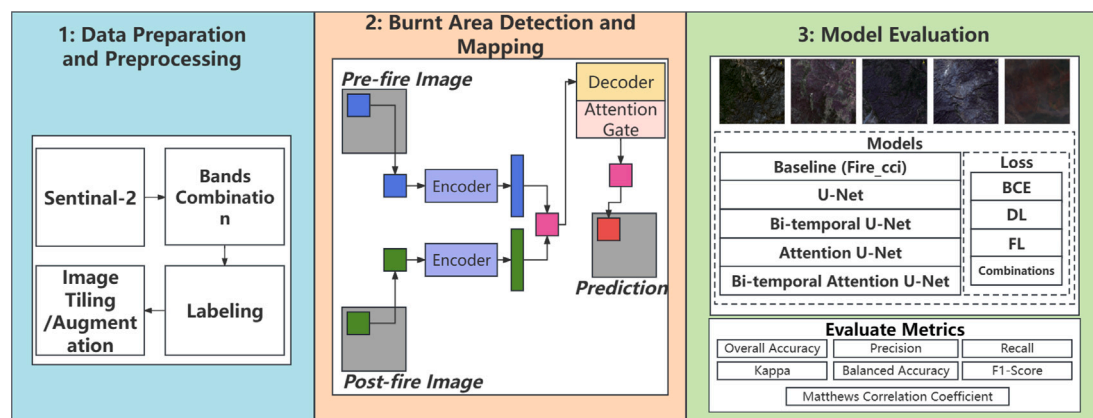


Fig. 4. Framework of burnt area detection and mapping using the bi-temporal attention U-Net model.

The Stochastic Gradient Descent (SGD) optimizer was selected for optimization. Three loss functions and their combinations, including BCE, DLoss, and Focal Loss (Floss), were examined and evaluated to determine the optimal model architecture for wildfire burnt area detection. In particular, BCE measures the difference between two probability distributions for a given random variable or set of events, and

is commonly applied for classification tasks. Given that segmentation involves pixel-level classification, BCE, defined as follows, works well for this task (Jadon, 2020).

$$BCE = -\frac{1}{N} \sum_{i=1}^N y_i \log(p_i) + (1 - y_i) \log(1 - p_i)$$

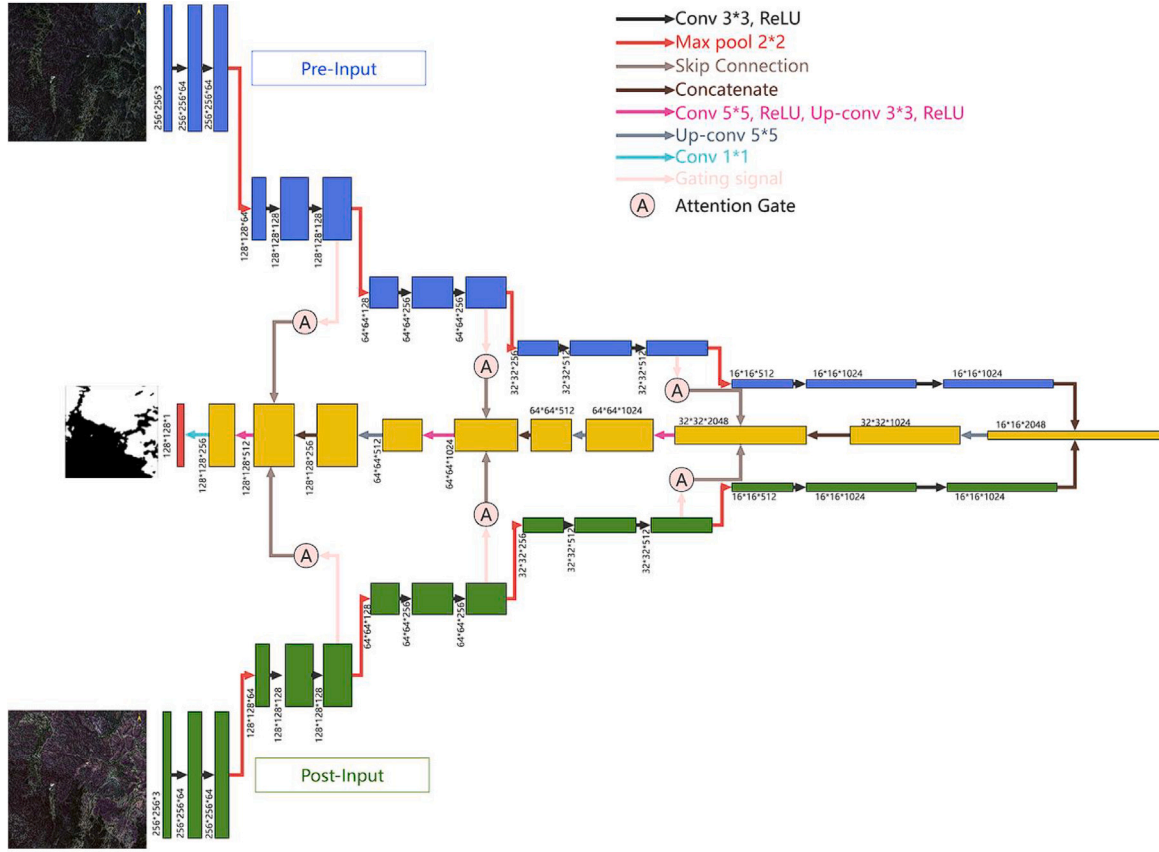


Fig. 5. Overview of the proposed BiAU-Net architecture for burnt area mapping.

where  $y_i \in \{0, 1\}$  is the ground truth label,  $p_i \in [0, 1]$  is the predicted probability.

DLoss quantifies the dissimilarity between predicted and true segmentation labels, aiming to identify specific objects or regions (i.e. burnt area) within an image. DLoss optimizes models by calculating the dice overlap coefficient between the ground truth and the prediction. This approach effectively addresses the imbalance between the foreground and background (Zhao et al., 2020). DLoss is defined as:

$$DLoss = 1 - \frac{2 \sum_{i=1}^N y_i p_i + smooth}{\sum_{i=1}^N y_i + \sum_{i=1}^N p_i + smooth}$$

where  $y_i \in \{0, 1\}$  is the ground truth,  $p_i \in [0, 1]$  denotes the predicted probability, and *smooth* is a smoothing factor to avoid division by zero.

FLoss is introduced to address the challenges arising from imbalanced datasets, characterized by one class largely outnumbering the other. In such cases, standard loss functions like BCE may yield sub-optimal results, as the model could exhibit bias towards the majority class and underperform on the minority class. To rectify this class imbalance issue, FLoss down-weights easy-to-classify examples and places more emphasis on hard-to-classify examples by incorporating two key parameters:  $\alpha$ , a weight parameter to balance between positive and negative samples, and  $\gamma$ , a tunable focusing parameter for adjusting the magnitude of the loss for well-classified examples based on the BCE. The formula for FLoss is given by:

$$FLoss = -\frac{1}{N} \sum_{i=1}^N \alpha_i (1 - p_i)^\gamma y_i \log(p_i) + (1 - \alpha_i) p_i^\gamma (1 - y_i) \log(1 - p_i)$$

where  $y_i \in \{0, 1\}$  represents the ground truth label, and  $p_i \in [0, 1]$  indicates the predicted probability.

### 3.2.3. Evaluation metrics

To assess the model's effectiveness, seven evaluation metrics were used to analyze the mapping results, including Overall Accuracy (OA),

Precision, Recall, F1-score, Balanced Accuracy (BA), Kappa Coefficient (KC), and Matthews Correlation Coefficient (MCC) (Csurka et al., 2013). These metrics provide valuable insights into the model's effectiveness. Given true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN) for model predication, the definition of each indicator is provided as below:

Overall accuracy (OA): OA quantifies the proportion of correctly predicted samples over all samples.

$$OA = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Precision gauges the proportion of correct positive samples overall positive samples.

$$Precision = \frac{TP}{TP + FP}$$

Recall: Recall measures the proportion of correct positive samples overall actual positive samples.

$$Recall = \frac{TP}{TP + FN}$$

F1-score: The F1-score, representing the harmonic mean of precision and recall, provides a balanced assessment between these two metrics.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Balanced accuracy (BA): BA denotes the average of sensitivity (true positive rate) and specificity (true negative rate).

$$BA = \frac{1}{2} \left( \frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)$$

Kappa coefficient (KC): KC assesses the agreement between predicted and actual classifications, accounting for the agreement expected by chance.

$$KC = \frac{p_o - p_e}{1 - p_e}$$

where  $p_o$  denotes the observed agreement, and  $p_e$  represents the expected agreement by chance.

Matthews Correlation Coefficient (MCC): MCC measures the quality of binary classifications.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

## 4. Experiments and results

### 4.1. Experimental design

The training dataset included 278 pairs of pre- and post-fire images captured in Chico, California with each image including a size of  $256 \times 256$  pixels. The training and validation datasets were randomly sampled from the California dataset in a 10:1 ratio. After 50 training epochs, the model with the minimum loss and maximum accuracy was selected for testing the California dataset and the datasets from four other continents. The confusion matrix generated from the test results was used to produce prediction images within the “Python-3.6” platform.

The experimental design in this study followed an ablation study approach. It involved a comparative assessment of different models, including the baseline (Fire\_cci51), U-Net, Bi-temporal U-Net, and Bi-temporal Attention U-Net, with various loss functions. The U-Net used BCE as the loss function, BiU-Net and BiAU-Net used BCE, DLoss, FLoss and their combinations as the loss function. The training data was sourced from one wildfire event in Chico, and the models were tested on Chico and four other events that occurred on different continents to evaluate the generalizability of the models. To assess the efficacy of the proposed models, we compared the predictions generated by the BiAU-Net and U-Net. This comparative analysis allowed us to determine the extent of improvement achieved through the incorporation of bi-temporal input, attention gates, and the specified loss function. To assess the model's effectiveness, seven evaluation metrics were used to analyze the mapping results, including OA, Precision, Recall, F1-score, KC, and MCC (Csurka et al., 2013). These metrics provide valuable insights into the model's effectiveness.

### 4.2. Results

This section presents the performance of each model (Table 1, Fig. 6, Fig. 7). For all test areas except for Sumatra Island, the U-Net model showed notably poorer performance compared to the baseline. However, when incorporating Bi-temporal input (BiU-Net with BCE as loss function), a significant improvement was observed, surpassing the baseline. Furthermore, when implementing the DLoss+FLoss loss function, the BA score exhibited a remarkable improvement (Table 1). For example, the BiU-Net achieved a BA score of 0.930, while the BiAU-Net surpassed with a BA score of 0.965 for Chico wildfire. The BA score serves as an indicator of the model's ability to handle imbalanced data. The FLoss increases the weight of misclassified samples and difficult-to-classify samples in the loss function, which biases the loss function towards challenging samples. These two features help improve the accuracy of these challenging samples. Ultimately, this leads to improved performance when dealing with imbalanced datasets. Given that the training dataset contains approximately 70% easy cases, characterized by small, contiguous burnt areas and most of them are gathered in one piece, and 30% complex cases, featuring large, distributed burnt areas, FLoss demonstrates superior effectiveness compared to BCE and DLoss due to its optimization for handling the imbalance between easy and complex cases. Note that FLoss tends to generate more FP, but such outcomes are acceptable within the context of the burnt area detection and mapping.

Interestingly, the implementation of attention gates alone did not improve the model for Chico and Bredbo cases. In particular, regarding metrics, such as KC and MCC, the BiU-Net model with attention

gates (BiAU-Net) performed less effectively than the version without attention gates (BiU-Net). This may be attributed to imbalances in the training data. However, attention gates did improve other metrics while only slightly diminishing KC and MCC. Furthermore, this deficiency could be mitigated by using the DLoss+FLoss loss function, which indeed led to improvements in all evaluated metrics. Below we elaborate several unique patterns across different case studies.

#### 4.2.1. Test area: Chico, California

In Chico, California, all models, including the baseline, demonstrated commendable performance as both the test and training data were derived from the same wildfire event. Fig. 7 presents the predictions generated by both the U-Net and BiAU-Net using the DLoss+FLoss loss function for Chico wildfire. In Fig. 3, a noteworthy observation emerges as both the U-Net and BiAU-Net falsely identified the north bareland as burnt area. This misclassification mainly occurred in two situations: (1) bareland and burnt areas appear visually similar in false color images. During wildfires, sand and soil are often used as protective barriers, and post-combustion material covers the bareland, making it difficult to distinguish bareland near burnt areas; and (2) the boundary of burnt areas, where there are mixed burnt and non-burnt regions. However, the BiAU-Net improved results in the central small burnt areas and at the edge of the wildfires by accurately identifying the non-burnt regions. This indicates its superior predictive capabilities within small areas where burnt areas and non-burnt areas are intermingled, showcasing enhanced performance in such complex scenarios.

#### 4.2.2. Test area: Bredbo, Australia

Fig. 7 shows that BiAU-Net outperformed U-Net in mapping details, especially for small non-burnt areas within large burnt areas. At the edge of the middle of large burnt areas and southwest of small burnt areas, U-Net produced a substantial false alarm, whereas BiAU-Net notably rectified this issue. Both U-Net and BiAU-Net falsely classified many backgrounds as burnt areas, similar to the misclassification observed in the California dataset. These misclassified regions do not constitute actual burnt areas, but are affected by nearby burnt areas.

Examining Fig. 3, it becomes apparent that the wildfire scenario and terrain in Bredbo are simpler than those in California, with both regions sharing quite similar forest environments. Consequently, the U-Net performed better in Bredbo than it did in California. Similarly to the case in California, the inclusion of attention gates alone could not improve the model and could potentially worsen it if the model was already satisfactory. However, by implementing DLoss+FLoss to help the model handle imbalanced data, all metrics were improved.

#### 4.2.3. Test area: Tarragona, Spain

It is evident the U-Net outperformed in Tarragona than it did in Bredbo (Table 1). However, by incorporating bi-temporal input, the BiU-Net model performed even worse than it did in Bredbo. The addition of attention gates brought the model's performance in Tarragona to a level similar to that observed in Bredbo. This implies that bi-temporal inputs are less beneficial and attention gates help more in Tarragona than in Bredbo. These can be explained by pre- and post-fire satellite images (Fig. 3), where there was a large area in the middle bottom and a relatively small area in the middle top that transitioned from a shallow green in the pre-fire image to brown in the post-fire image, which is distinct from the predominant change observed in Chico, where the transition is from bright blue to brown. This substantial difference disrupts the training process when using bi-temporal inputs. However, the addition of attention gates directs the model's focus towards changes similar to those in Chico, mitigating the impact of the divergent changes observed in Tarragona. This is why attention gates bring the model's performance to a level similar to Bredbo.

BiAU-Net shows improvement, particularly in the central bareland and at the edge of the bareland adjacent to the southern forests. Unlike

**Table 1**  
Performance comparison of different models.

| Case           | Index     | Baseline     | U-Net        | BiU-Net<br>(BCE) | BiU-Net<br>(DLoss+FLoss) | BiAU-Net<br>(BCE) | BiAU-Net<br>(DLoss+FLoss) |
|----------------|-----------|--------------|--------------|------------------|--------------------------|-------------------|---------------------------|
| Chico          | OA        | 0.892        | 0.823        | 0.928            | <b>0.931</b>             | 0.929             | <b>0.968</b>              |
|                | Precision | 0.923        | 0.850        | 0.965            | 0.929                    | <b>0.966</b>      | <b>0.974</b>              |
|                | Recall    | 0.888        | 0.754        | <b>0.939</b>     | 0.921                    | <b>0.939</b>      | <b>0.949</b>              |
|                | F1-score  | 0.910        | 0.799        | <b>0.952</b>     | 0.925                    | <b>0.952</b>      | <b>0.961</b>              |
|                | BA        | 0.655        | 0.819        | 0.768            | <b>0.930</b>             | 0.775             | <b>0.965</b>              |
|                | KC        | 0.630        | 0.643        | 0.762            | <b>0.863</b>             | 0.714             | <b>0.934</b>              |
|                | MCC       | 0.655        | 0.647        | 0.768            | <b>0.858</b>             | 0.716             | <b>0.929</b>              |
| Bredbo         | OA        | 0.828        | 0.785        | 0.923            | <b>0.927</b>             | 0.921             | <b>0.941</b>              |
|                | Precision | 0.556        | 0.737        | 0.827            | <b>0.928</b>             | 0.832             | <b>0.951</b>              |
|                | Recall    | 0.869        | <b>0.940</b> | <b>0.953</b>     | 0.920                    | <b>0.953</b>      | 0.926                     |
|                | F1-score  | 0.678        | 0.827        | 0.885            | <b>0.924</b>             | 0.887             | <b>0.939</b>              |
|                | BA        | 0.843        | 0.714        | <b>0.931</b>     | 0.927                    | 0.929             | <b>0.946</b>              |
|                | KC        | 0.569        | 0.428        | 0.828            | <b>0.852</b>             | 0.828             | <b>0.895</b>              |
|                | MCC       | 0.595        | 0.433        | 0.833            | <b>0.859</b>             | 0.833             | <b>0.889</b>              |
| Tarragona      | OA        | 0.879        | 0.866        | 0.909            | 0.913                    | <b>0.927</b>      | <b>0.939</b>              |
|                | Precision | 0.748        | 0.712        | 0.826            | <b>0.943</b>             | 0.849             | <b>0.909</b>              |
|                | Recall    | 0.851        | 0.820        | 0.900            | 0.879                    | <b>0.947</b>      | <b>0.944</b>              |
|                | F1-score  | 0.796        | 0.762        | 0.861            | <b>0.910</b>             | 0.895             | <b>0.926</b>              |
|                | BA        | 0.870        | 0.851        | 0.907            | 0.913                    | <b>0.932</b>      | <b>0.940</b>              |
|                | KC        | 0.710        | 0.670        | 0.794            | 0.826                    | <b>0.840</b>      | <b>0.878</b>              |
|                | MCC       | 0.713        | 0.673        | 0.796            | 0.828                    | <b>0.843</b>      | <b>0.876</b>              |
| Sumatra Island | OA        | 0.709        | 0.810        | 0.912            | <b>0.919</b>             | <b>0.920</b>      | 0.904                     |
|                | Precision | 0.633        | 0.794        | 0.839            | <b>0.888</b>             | 0.879             | <b>0.910</b>              |
|                | Recall    | 0.719        | 0.869        | 0.922            | <b>0.934</b>             | <b>0.948</b>      | 0.927                     |
|                | F1-score  | 0.684        | 0.834        | 0.840            | <b>0.910</b>             | 0.865             | <b>0.931</b>              |
|                | BA        | 0.739        | 0.900        | 0.909            | <b>0.920</b>             | 0.902             | <b>0.922</b>              |
|                | KC        | 0.482        | 0.759        | 0.814            | <b>0.836</b>             | 0.833             | <b>0.845</b>              |
|                | MCC       | 0.479        | 0.768        | 0.818            | 0.837                    | <b>0.839</b>      | <b>0.891</b>              |
| Nairobi        | OA        | 0.682        | 0.610        | 0.746            | <b>0.777</b>             | 0.765             | <b>0.816</b>              |
|                | Precision | 0.355        | 0.865        | <b>0.925</b>     | 0.834                    | <b>0.933</b>      | <b>0.925</b>              |
|                | Recall    | <b>0.776</b> | 0.281        | 0.546            | 0.682                    | 0.576             | <b>0.694</b>              |
|                | F1-score  | 0.487        | 0.424        | 0.687            | <b>0.750</b>             | 0.712             | <b>0.793</b>              |
|                | BA        | 0.712        | 0.617        | 0.750            | <b>0.775</b>             | 0.767             | <b>0.818</b>              |
|                | KC        | 0.300        | 0.232        | 0.496            | <b>0.553</b>             | <b>0.533</b>      | <b>0.634</b>              |
|                | MCC       | 0.348        | 0.316        | 0.545            | 0.562                    | <b>0.577</b>      | <b>0.620</b>              |

\*Each evaluation metric is averaged ten times.

the bareland in California, which remains distinguishable from burnt areas after wildfires, the bareland in Tarragona exhibits greater similarity to burnt areas after wildfires. This may be primarily due to the presence of small groups of combustible matter such as fallen leaves, low-density weeds, and dead wood in the bareland of Tarragona. Before wildfires, the spectrum of bareland in Tarragona and California was very similar because the density of combustible matter in Tarragona is so low that it has a minimal impact on the overall spectrum. However, during wildfires in Tarragona, the combustible matter ignites, leaving behind ashes and other debris that cover the bareland and creating a noticeable difference with California's bareland, which has minimal combustible material to burn.

The attention gates caused the model to focus more on burnt areas. Therefore, even if the bareland after a wildfire resembled burnt areas in terms of spectral characteristics, the model still focused on larger burnt areas instead of small areas where burnt areas and bareland mixed up. In addition, the attention gates highlighted the salient features of a burnt area, aiding in the identification of subtle differences between burnt area and bareland after wildfires.

#### 4.2.4. Test area: Sumatra island, Indonesia

Sumatra Island exhibited a different pattern in the false color satellite images compared to California, Bredbo, and Tarragona areas, even though it was covered by vegetation (Fig. 3). Instead of the blue appearance associated with vegetation in the aforementioned regions, the vegetation in Sumatra Island appeared as a shallow green before the wildfire. However, there are two main similarities in the change tendency of false color images before and after the wildfire between Sumatra Island and Chico. First, the change is remarkable in both locations. Second, although the burnt areas in false color images in

Chico are mostly deep blue, and in Sumatra Island are shallow green and shallow blue in pre-fire images, the post-fire burnt area appears as brown regardless of its depth. Therefore, by implementing bi-temporal images, the change can be utilized to extract features that can assist the model in handling the Sumatra Island case. The introduction of bi-temporal input resulted in a significant improvement from U-Net to BiU-Net. In addition, the implementation of attention gates helps the model to focus more on the burnt area, which is more significant for the training process, while effectively reducing the impact on the model caused by less important areas (non-burnt areas) and even noise. Therefore, implementing attention gates with a suitable loss function greatly improved the model's performance (Fig. 7). Notably, BiAU-Net made substantial improvements in the central non-burnt area, where U-Net tended to incorrectly identify non-burnt area as burnt area.

#### 4.2.5. Test area: Nairobi, Kenya

The temporal resolution of Sentinel-2 MSI 2A products in Nairobi, Kenya is notably lower at 5 days compared to other areas, where it can achieve a resolution of 2 days in locations such as Chico, Bredbo, and Tarragona. A higher temporal resolution provides more options for selecting satellite images with good sunlight and minimal cloud coverage. While both Nairobi and Sumatra Island only have a 5-day temporal resolution, Sumatra Island benefits from more favorable weather conditions for capturing images with Sentinel-2. As shown in Table 1, the baseline, based on MODIS data, also performed poorly. Additionally, the environmental conditions in this area are significantly different from California; As highlighted in Fig. 3, the primary soil type, red soil in western Kenya, displays distinct spectral characteristics compared to California and the other study areas, both in true color and false color images. Despite these differences, the burnt areas still appear

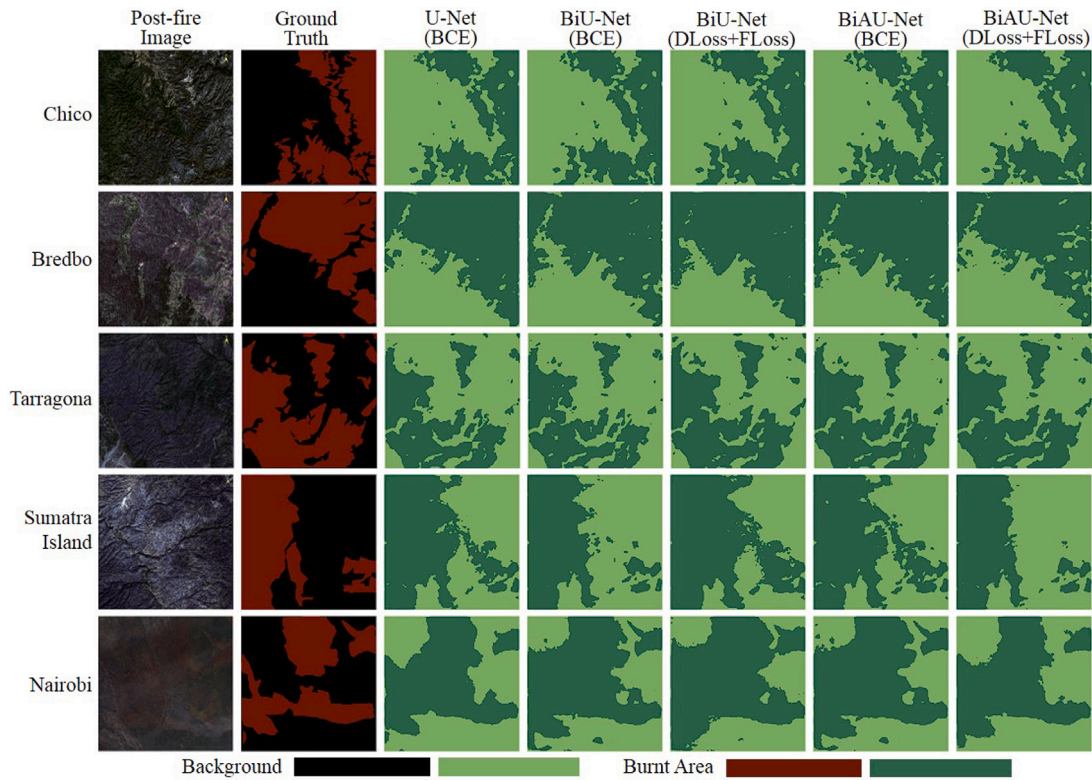


Fig. 6. The prediction of different models in five study areas.

as deep brown in false color images, suggesting that the models have the ability to handle the Nairobi case.

Fig. 7 shows that both U-Net and BiAU-Net struggled in identifying non-burnt regions between two large burnt areas. The challenge arises from the different spectral characteristics of the Nairobi test area compared to the training area in California. Notably, the incorporation of bi-temporal input improved the U-Net by integrating the changes of pre- and post-fire images as features. Although these features differed from those in the training area, they aided the model in correctly identifying the regions with changes (i.e. being burnt). The utilization of attention gates, coupled with the use of DLoss and FLoss as the loss function mainly improved the detection in the edge areas.

## 5. Discussion

### 5.1. Impact of model architecture and design

Fig. 8 displays the ROC curves of different models in Chico. The False Positive Rate (FPR) and True Positive Rate (TPR) can indicate the sensitivity and specificity of the models. Even if the distribution of positive and negative samples changes, the shape of the curve can remain largely unchanged. Therefore, this evaluation metric can reduce the interference caused by different test sets and provide a more objective measure of the model's performance. The Area Under the ROC Curve (AUC), which ranges from 0.1 to 1, can be used as a value to intuitively assess the quality of the classifier. The larger the value, the better the performance.

When the test area exhibits significant similarities with the training area, U-Net generally performs adequately. However, the introduction of bi-temporal input can further enhance the performance of the U-Net based models by considering the changes between pre- and post-fire images in both burnt and non-burnt regions. This improvement becomes more evident when the test area's environment significantly differs in spectral features from the training area. When incorporating attention gates, it is important to design an appropriate loss function.

Table 2

Performance of different loss functions in Chico.

| Model    | Index     | BCE          | BCE+DLoss | BCE+FLoss    | DL+FLoss     |
|----------|-----------|--------------|-----------|--------------|--------------|
| BiU-Net  | OA        | <b>0.928</b> | 0.901     | 0.912        | <b>0.931</b> |
|          | Precision | <b>0.965</b> | 0.927     | 0.907        | <b>0.929</b> |
|          | Recall    | <b>0.939</b> | 0.854     | 0.918        | <b>0.921</b> |
|          | F1-score  | <b>0.952</b> | 0.889     | 0.913        | <b>0.925</b> |
|          | BA        | 0.768        | 0.898     | <b>0.912</b> | <b>0.930</b> |
|          | KC        | 0.762        | 0.811     | <b>0.819</b> | <b>0.863</b> |
|          | MCC       | 0.768        | 0.803     | <b>0.826</b> | <b>0.858</b> |
| BiAU-Net | OA        | <b>0.929</b> | 0.921     | 0.923        | <b>0.968</b> |
|          | Precision | <b>0.966</b> | 0.963     | 0.875        | <b>0.974</b> |
|          | Recall    | 0.939        | 0.937     | <b>0.967</b> | <b>0.949</b> |
|          | F1-score  | <b>0.952</b> | 0.950     | 0.926        | <b>0.961</b> |
|          | BA        | 0.775        | 0.782     | <b>0.911</b> | <b>0.965</b> |
|          | KC        | 0.714        | 0.719     | <b>0.847</b> | <b>0.934</b> |
|          | MCC       | 0.716        | 0.721     | <b>0.851</b> | <b>0.929</b> |

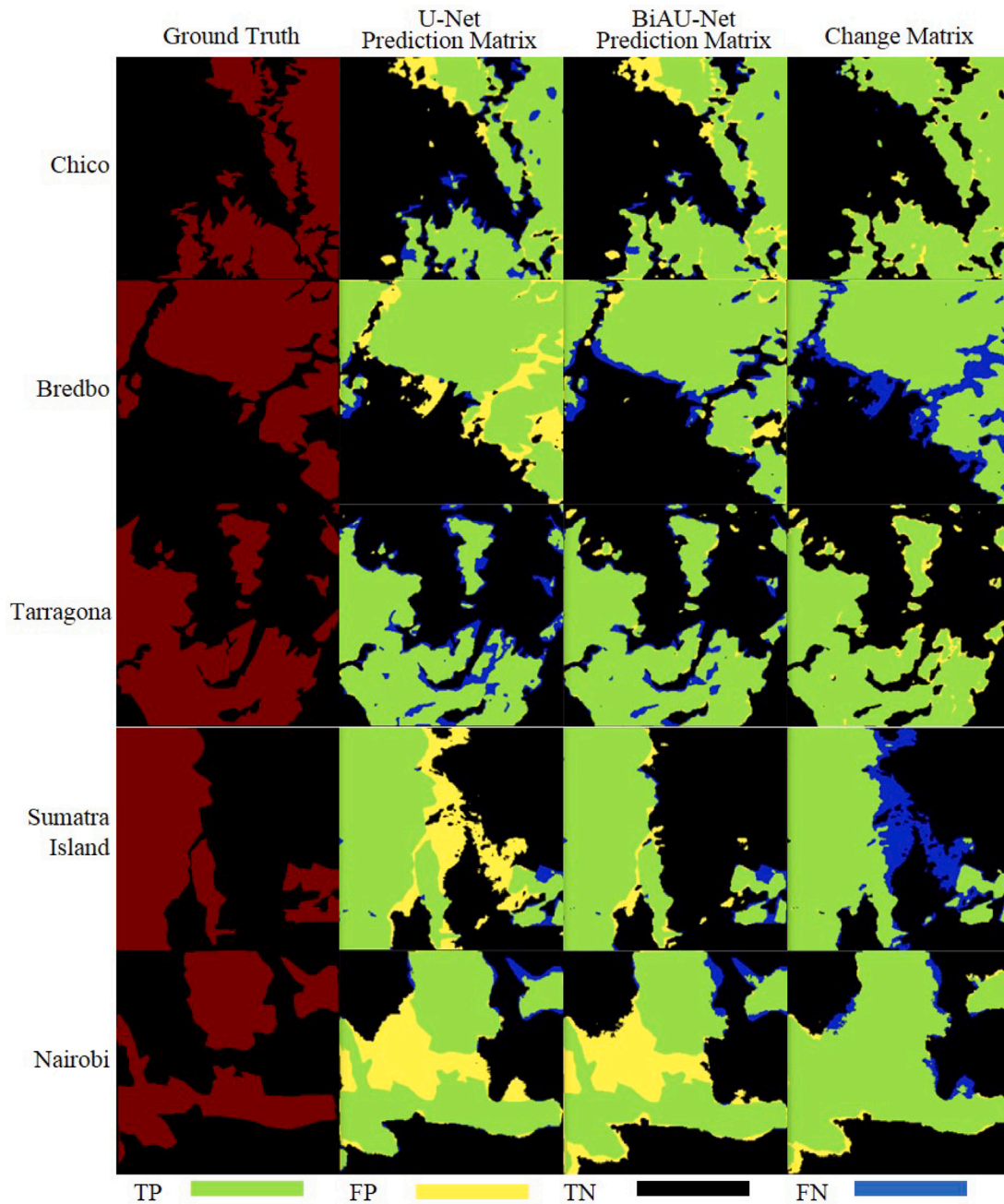
\*Each evaluate metric is averaged ten times.

In particular, BCE alone may not yield satisfactory results and could potentially degrade model performance due to dataset imbalances. However, switching to a combination of BCE and DLoss as a loss function can provide some improvement, as DLoss helps address imbalanced classes. The most effective combination appears to be DLoss and FLoss, as FLoss specifically addresses the imbalance between foreground (burnt) and background (non-burnt) areas. This configuration yields the best model performance, particularly in areas where burnt and non-burnt regions are mixed and at the edges of burnt areas and non-burnt areas.

### 5.2. Loss function and attention

#### 5.2.1. Loss function

The dataset exhibits two dimensions of imbalance: (1) imbalance between foreground (burnt) and background (non-burnt) within one



**Fig. 7.** Confusion matrices of predictions and changes of U-Net (BCE) and BiAU-Net (DLoss+FLoss) in five study areas; note the prediction mask (second and third column) uses colors: green for true positive (TP), black for true negative (TN), yellow for false positive (FP), and blue for false negative (FN). In the comparison between U-Net and the proposed BiAU-Net (fourth column), yellow indicates BiAU-Net's conversion of U-Net's FN to TP or TN, while blue indicates BiAU-Net's conversion of U-Net's FP to TP or TN.

training dataset (image); and (2) imbalance in the complexity of different training datasets (samples) with varying level of challenges for burnt area segmentation, and therefore contribute to model learning differently. To address these imbalances, we experimented with different loss functions, including DLoss, FLoss, BCE, and various combinations of these. We employed the Chico case study to design and evaluate the performance of these different loss functions, as presented in Table 2. These functions are applied to the BiU-Net model, both with and without the attention mechanism. The loss functions under consideration are BCE, BCE+DLoss, BCE+FLoss, and DLoss+FLoss, with their definitions detailed in Section 3.2.2.

First, DLoss demonstrates commendable performance when dealing with scenarios characterized by severe imbalances between foreground and background (Wang et al., 2020). DLoss makes the model focus

more on the foreground during the training process, especially on smaller objects. This is because DLoss uses a region-related loss function. It ensures that the loss of one pixel is not only related to that pixel but also to other pixels. No matter how large the image is, the losses of the same area of the foreground remain consistent, and the contribution of the same area of the foreground does not change with the size of the image (Zhao et al., 2020). Therefore, when the foreground has less proportion, more background is merged to maintain a balanced ratio between the foreground and background. DLoss excels in mining the foreground and performs better in small areas where burnt and not-burnt areas are mixed up. However, DLoss may lead to gradient saturation in certain cases, especially when dealing with small foreground regions. While DLoss performs well in these regions, it requires a few misclassifications as a precondition. If too many

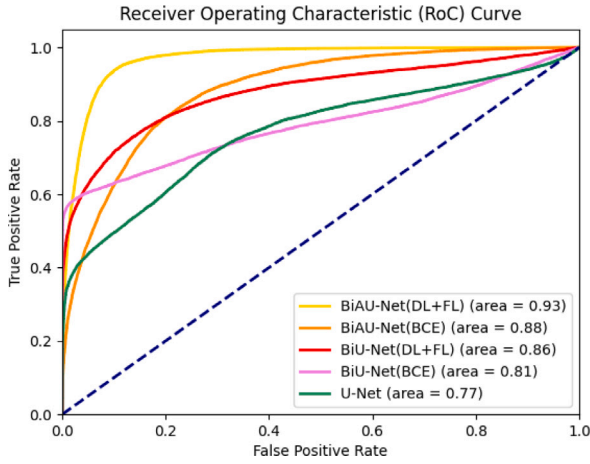


Fig. 8. RoC curves of different models in Chico.

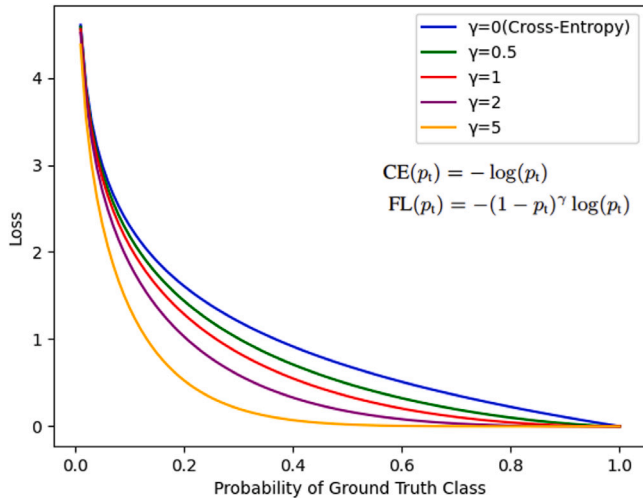


Fig. 9. Losses of cross-entropy and focal loss with different focusing parameter  $\gamma$ .

pixels are incorrectly classified, it can lead to model divergence (Li et al., 2019). This can explain why, in the BiU-Net model, although the combination of BCE and DLoss compared to BCE shows improvement in terms of the BA, KC, and MCC, the OA, Precision, Recall, and F1-score are worse.

Second, imbalanced complexity of different datasets pose a challenge when attempting to classify categories with limited training samples. Therefore, by focusing on the difficult samples, the FLoss addresses the issue of low accuracy in classifying categories with fewer samples. It is important to note that difficult samples are not only caused by fewer samples. In essence, FLoss not only mitigates the problem of sample imbalance but also enhances the overall model performance. FLoss provides the models with more confidence in their predictions. The models can tolerate certain misclassifications in easy samples by decreasing their weight. As a result, the models can take more risks. This is particularly important when dealing with imbalanced datasets, especially in burnt area detection, where the easy samples are mostly background. In this case, a few FPs can be acceptable (Hossain et al., 2021).

When handling an imbalanced dataset, particularly in object detection, the majority of pixels represent the background (non-burnt), while only a minority correspond to the foreground of interest. FLoss dynamically adjusts focus, with the focusing parameter  $\gamma$  decreasing as the confidence coefficient rises. This reduces the weight given to

easier examples and shifts the model's focus towards more complex ones (Lin et al., 2017). In Fig. 9, when  $\gamma$  is higher than 1, the focal loss can decrease the loss for examples with a high probability of the ground truth class, in other words, easy examples (Doi and Iwasaki, 2018). However, FLoss may lead to poor generalization of models, which clarifies why in the BiU-Net model, although the combination of BCE and FLoss performs much better in BA, KC, and MCC compared to the BCE model, the OA, Precision, Recall, and F1-score are worse, similar to BCE+DLoss. Nonetheless, when combining DLoss and FLoss, the model performance and stability are greatly improved, surpassing other loss functions. This combination is a common practice in the field of image segmentation field (Yeung et al., 2022).

### 5.2.2. Attention mechanism

The attention mechanism improves the model's capability to concentrate on crucial features across various scales of the feature map. It serves as a filter, effectively suppressing irrelevant areas during the encoding process. Notably, this mechanism does not require additional labels, only a few extra calculations. In the Chico case study, the BiU-Net with attention gates improved the performance comprehensively (Table 2). While the enhancement using attention with BCE loss function was marginal, incorporating DLoss or FLoss amplified the improvement, culminating in optimal performance when DLoss and FLoss were combined. Attention Gates enable the model to prioritize important features and measure the importance of features by weights (Gao et al., 2022). By dynamically assessing the relevance of the features to create attention weights, and applying these weights to the original features, useful features are strengthened, and irrelevant ones are de-emphasized, therefore enhancing model efficiency and performance (Oktay et al., 2018).

Fig. 10 shows the visualization of attention weights for a training tile. The attention gates allow the model to emphasize boundaries and small areas by amplifying the attention weights. This process intensifies with the progression of training epochs. To sum up, the attention mechanism has two advantages: (1) making the model focus on the useful features; and (2) offering an explanation of the forward and backward propagation process in DL training (Monaco et al., 2021).

### 5.3. Impact of study area

Variations in spectral characteristics within different environments contribute to variations in model performance across different regions. This study examines two scenarios: (1) training and test areas share certain similarity in terms of environmental conditions, and (2) training and test areas with distinct environmental conditions. In the first scenario, the test areas, including Tarragona and Bredbo, share similar environmental conditions (e.g., vegetation density) reflected in the images, with the training area, Chico, California. While the burnt or non-burnt areas in the pre-fire or post-fire images closely resemble the training area, one of the bi-temporal images has different spectral characteristics. The use of bi-temporal input allows the model to incorporate the same spectral features from pre- or post-fire images, where the burnt or the non-burnt areas are likely in different areas, into the model, enabling the model to operate in different areas.

In the second scenario, both burnt areas and non-burnt areas in pre- and post-fire images, such as Nairobi, significantly differ from those in the training area (i.e., Chico, California). However, the change of spectral features caused by wildfires can be captured as features by using bi-temporal input. In particular, the two major changes caused by wildfires include noticeable changes in spectral features and the distinct brown hue of burnt areas in post-fire images. The model can rely on these features to compensate for the slight similarity of satellite images in different areas.

While assessing the model's adaptability to different areas, we experimented with applying histogram matching to preprocess the test images, using satellite images from Chico as references, for model

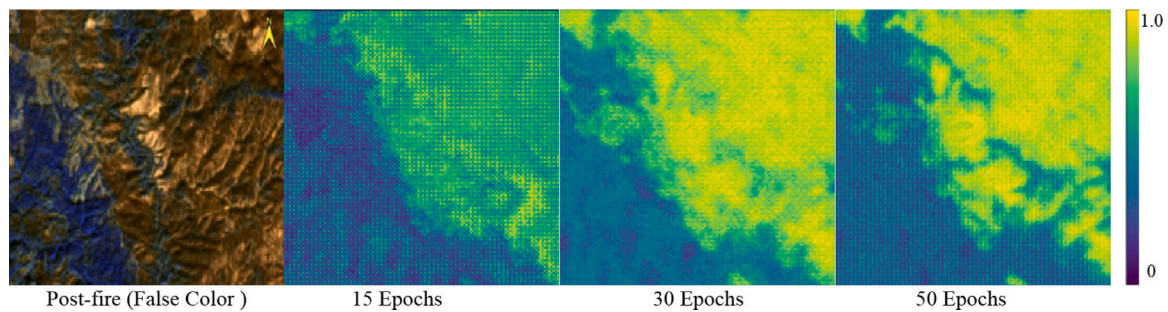


Fig. 10. Visualization of attention weights in Chico.

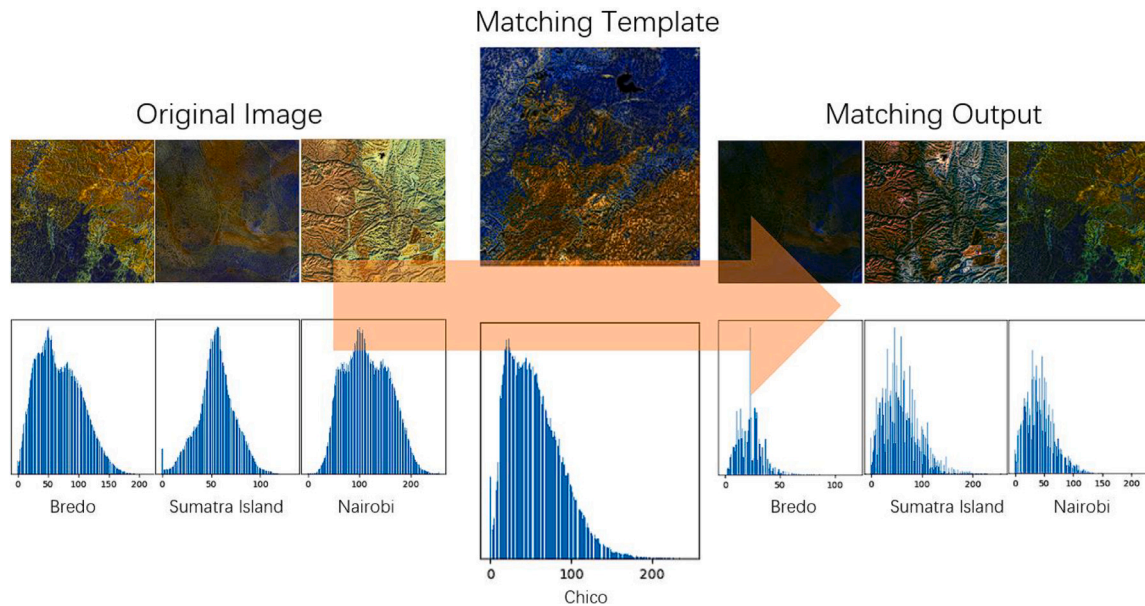


Fig. 11. Applying histogram match in Bredbo, Sumatra Island and Nairobi.

training, as depicted in Fig. 11. These adjusted pre- and post-fire images were subsequently fed into BiAU-Net to evaluate the outcomes. The results were found to be highly dependent on the images of test areas. As previously discussed, the inherent dataset imbalance had a profound impact on the results. This imbalance is multifaceted, affected not only by the ratio of foreground to background but also by variations in spectral features within both foreground and background regions. As depicted in Fig. 12, the results after histogram matching were even worse than without it in Bredbo and Sumatra Island. While it improved the results in Nairobi by reducing the FP, it simultaneously increased the FN.

#### 5.4. Performance comparison with prior models

Different ML based models have been proposed for burn area detection and mapping with RS imagery in recent years. Table 3 presents datasets and results of some well-established models alongside our proposed BiAU-Net. While all models performed satisfactorily, the proposed model stands out due to its superior generalizability. It was evaluated on datasets with the largest spatial distance and in different continents with highly different geographic and environment contexts. The models with higher OA and F1-Score, such as QMDNN-Net (Seydi et al., 2022b), BAMUUPI (Cho et al., 2022), and CST (Seydi and Sadegh, 2023), were tested and applied in the same training regions.

Moreover, while the proposed model do not achieve the highest OA, its OA remains relatively high. This indicates a greater number of TP and TN compared to FP and FN, suggesting that the proposed

model correctly predicts burnt and non-burnt areas. The proposed model also achieves the best Precision in test areas from the US, Spain, and Australia. High Precision indicates that most predicted burnt areas are correctly identified, with fewer non-burnt areas incorrectly classified as burnt. Additionally, the proposed model has a high F1-score compared to Burnt-Net (Seydi et al., 2022a), which represents effective performance in both Precision and Recall. This indicates that the proposed model not only accurately predicts burnt areas but also ensures these predictions are reliable. The relatively high KC and MCC further validate this, indicating a good agreement with the ground truth.

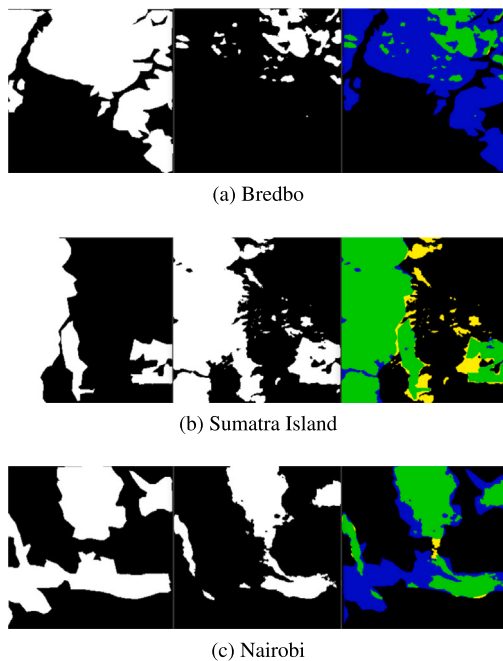
Popular global wildfire burnt area products, like Fire\_cci51 and MODIS MCD64A1, are frequently used to examine wildfire dynamics, evaluate fire impacts on ecosystems, and improve fire forecasts (Chen et al., 2023). However, they have limitations in spatial resolution - 250 m and 500 m respectively. This can cause the product to often overlook small fires, leading to an underestimation of wildfires (Jones et al., 2022). This may explain why Fire\_cci51's performance falls short in the five study areas (Table 1). Our proposed method offers not just heightened accuracy, but also a finer spatial resolution of 20 m. In addition, although our model was trained and tested with Sentinel-2 images, it has the flexibility to handle inputs from a wide range of sources beyond just Sentinel-2.

#### 6. Conclusion and future work

This paper presents BiAU-Net, a CNN based on the U-Net architecture, designed for the identification and mapping of burnt areas using

**Table 3**  
Performance evaluation with prior models.

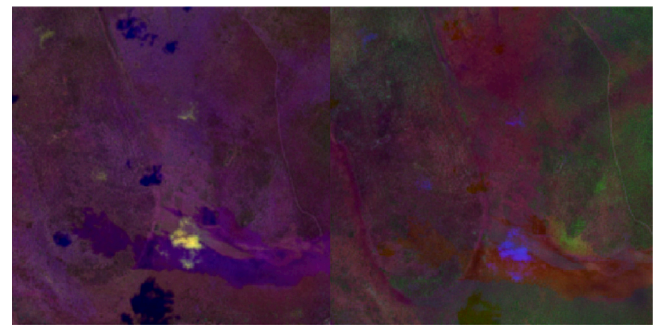
| Model name                      | Train dataset                   | Test dataset                                         | OA    | Precision | F1-score | KC    | MCC   |
|---------------------------------|---------------------------------|------------------------------------------------------|-------|-----------|----------|-------|-------|
| BIAU-Net (Proposed)             | the US                          | the US & Spain & Australia & Indonesia & Kenya       | 0.913 | 0.933     | 0.910    | 0.837 | 0.841 |
|                                 |                                 | the US & Spain & Australia                           | 0.949 | 0.956     | 0.942    | 0.902 | 0.929 |
| Burnt-Net (Seydi et al., 2022a) | Greece & France & Spain         | Cyprus & Turkey & Greece & France & Portugal & Spain | 0.968 | 0.931     | 0.926    | 0.905 | 0.908 |
| NRTWPM (Ban et al., 2020)       | North America                   | North America                                        | 0.888 | 0.798     | 0.875    | 0.776 | N/A   |
| QMDNN-Net (Seydi et al., 2022b) | Turkey & Greece                 | Turkey & Greece                                      | 0.961 | 0.950     | 0.960    | 0.894 | N/A   |
| BAMUUPI (Cho et al., 2022)      | Republic of Korea               | Republic of Korea                                    | 0.995 | N/A       | 0.903    | N/A   | N/A   |
| CST (Seydi and Sadegh, 2023)    | France & Greece & Spain & Chile | France & Greece & Spain                              | 0.980 | 0.952     | 0.959    | N/A   | 0.945 |
| U-Net (Knopp et al., 2020)      | Mostly Mediterranean            | Italy & Spain & UK                                   | 0.98  | 0.87      | 0.92     | 0.93  | N/A   |



**Fig. 12.** Prediction results after histogram matching in (a) Bredbo, (b) Sumatra Island and (c) Nairobi; From left to right: ground truth, histogram matching prediction, confusion matrix; green: true positives (burnt pixels); black: true negatives (non-burnt pixels); yellow: false positives; blue: false negatives.

satellite images from Sentinel-2. A comparative analysis is performed with the Fire\_cci51 product serving as a baseline and varying U-Net based model architecture. BiAU-Net boasts a remarkable model structure characterized by three essential components: bi-temporal input, attention gates, and tailored loss function. First, the bi-temporal input enhances the model's adaptability to diverse regions by utilizing pairs of pre- and post-fire images. Second, the incorporation of attention gates, coupled with a specifically crafted loss function that combines FLoss and DLoss loss functions, mitigates issues arising from imbalanced datasets, and enables the model to focus on boundaries and small areas where burn areas and non-burnt areas are mixed.

To test the model's generalizability, it is evaluated across five distinct testing areas with varying geographic and environmental contexts across different continents. In particular, these areas exhibit variations



**Fig. 13.** Pre- and post-fire Sentinel-2 MSI with SWIR (B12), SWIR (B11) and Narrow NIR (B8A) bands combination provide a clearer distinction between BA and non-BA areas in Nairobi.

in two key aspects: (1) different environmental contexts with different spectral characteristics (e.g., Nairobi); and (2) similar environmental context but with different spectral characteristics captured in the pre- and post-fire images (e.g., Tarragona and Sumatra Island). The experimental results indicate that (1) The use of bi-temporal input can contribute to the extraction of more useful features for model training. If the spectral features of the test area and training area differ in one temporal image, another temporal image can be used for feature extraction; and (2) The bi-temporal input and the model's architecture enable the model to consider the changes caused by wildfires as features. Even if there is no similarity in the spectrum between the test areas (e.g., Nairobi) and the training area (Chico, California) in both pre- and post-images, the trend and aspects of change can help the model in classification.

Future efforts are poised to expand and refine the capabilities of the BiAU-Net model for burnt area detection and mapping. One avenue of exploration involves the design and implementation of flexible attention gate locations based on the scale and spatial resolution of the datasets. Currently, attention gates are strategically placed at every up-sampling step. However, this may not always be optimal for different datasets. In later layers, the expansion of feature maps results in substantial size, yet the channel number remains constrained. The extracted channel weights exhibit a broad generality, lacking focus on specific features. Consequently, the extracted spatial attention may potentially have negative effects due to the limited channel capacity, absence of specificity, and the challenge of learning spatial attention.

Additionally, this study uses three bands (SWIR, SWIR and Narrow NIR) from Sentinel-2 images. However, this approach performed poorly in regions (e.g., Nairobi, Fig. 13) with significantly different environments from the training area (e.g., Chico). If we add more bands to the input and the model can assign weights to different channels, it may further help the model operate in foreign areas with varying geographic and environmental contexts. Moreover, this research used the Fire\_cci51 as the baseline to assess the performance of the proposed model. Future evaluations could also consider using alternative fire burnt area products, such as the fifth version of the Global Fire Emissions Database (GFED5; Chen et al., 2023).

### CRedit authorship contribution statement

**Tang Sui:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation. **Qunying Huang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Mingda Wu:** Writing – review & editing. **Meiliu Wu:** Writing – review & editing. **Zhou Zhang:** Writing – review & editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

The codes and the dataset are available on GitHub: <https://github.com/TangSui122/Bi-temporal-Wildfire-Burnt-Area-Detection.git>.

### Acknowledgments

This research is funded by the Office of the Vice Chancellor for Research and Graduate Education at the University of Wisconsin-Madison with funding from the Wisconsin Alumni Research Foundation, USA. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the funders.

### References

- Ban, Y., Zhang, P., Nascetti, A., Bevington, A.R., Wulder, M.A., 2020. Near real-time wildfire progression monitoring with Sentinel-1 SAR time series and deep learning. *Sci. Rep.* 10 (1), 1322.
- Bar, S., Parida, B.R., Pandey, A.C., 2020. Landsat-8 and Sentinel-2 based forest fire burn area mapping using machine learning algorithms on GEE cloud platform over Uttarakhand, Western Himalaya. *Remote Sens. Appl.: Soc. Environ.* 18, 100324.
- Bello, I., Zoph, B., Vaswani, A., Shlens, J., Le, Q.V., 2019. Attention augmented convolutional networks. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 3286–3295.
- Boer, M.M., Macfarlane, C., Norris, J., Sadler, R.J., Wallace, J., Grierson, P.F., 2008. Mapping burned areas and burn severity patterns in SW Australian eucalypt forest using remotely-sensed changes in leaf area index. *Remote Sens. Environ.* 112 (12), 4358–4369.
- Buslaev, A., Seferbekov, S., Iglovikov, V., Shvets, A., 2018. Fully convolutional network for automatic road extraction from satellite imagery. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. pp. 207–210.
- Chen, Y., Hall, J., Van Wees, D., Andela, N., Hantson, S., Giglio, L., Van Der Werf, G.R., Morton, D.C., Randerson, J.T., 2023. Multi-decadal trends and variability in burned area from the 5th version of the global fire emissions database (GFED5). *Earth Syst. Sci. Data Discussions* 2023, 1–52.
- Chen, W., Li, X., He, H., Wang, L., 2017. A review of fine-scale land use and land cover classification in open-pit mining areas by remote sensing techniques. *Remote Sens.* 10 (1), 15.
- Chen, T., Zheng, X., Niu, R., Plaza, A., 2022. Open-pit mine area mapping with gaofen-2 satellite images using U-Net+. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 15, 3589–3599.

- Cho, A.Y., Park, S.-e., Kim, D.-j., Kim, J., Li, C., Song, J., 2022. Burned area mapping using unitemporal planetscope imagery with a deep learning based approach. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 16, 242–253.
- Collins, L., Griffioen, P., Newell, G., Mellor, A., 2018. The utility of random forests for wildfire severity mapping. *Remote Sens. Environ.* 216, 374–384.
- Csurka, G., Larlus, D., Perronnin, F., Meylan, F., 2013. What is a good evaluation measure for semantic segmentation? In: *Bmvc*, vol. 27, (no. 2013), Bristol, pp. 10–5244.
- Cutler, D.R., Edwards, Jr., T.C., Beard, K.H., Cutler, A., Hess, K.T., Gibson, J., Lawler, J.J., 2007. Random forests for classification in ecology. *Ecology* 88 (11), 2783–2792.
- Doi, K., Iwasaki, A., 2018. The effect of focal loss in semantic segmentation of high resolution aerial image. In: *IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, pp. 6919–6922.
- Dragozi, E., Gitas, I.Z., Stavrakoudis, D.G., Theocharis, J.B., 2014. Burned area mapping using support vector machines and the FuzCoC feature selection method on VHR IKONOS imagery. *Remote Sens.* 6 (12), 12005–12036.
- Eidenshink, J., Schwind, B., Brewer, K., Zhu, Z.-L., Quayle, B., Howard, S., 2007. A project for monitoring trends in burn severity. *Fire Ecol.* 3 (1), 3–21.
- Escuina, S., Navarro, R., Fernández, P., 2008. Fire severity assessment by using NBR (Normalized Burn Ratio) and NDVI (Normalized Difference Vegetation Index) derived from LANDSAT TM/ETM images. *Int. J. Remote Sens.* 29 (4), 1053–1073.
- Eva, H., Lambin, E., 1998. Burnt area mapping in Central Africa using ATSR data. *Int. J. Remote Sens.* 19 (18), 3473–3497.
- Gao, Y., Guan, J., Zhang, F., Wang, X., Long, Z., 2022. Attention-unet-based near-real-time precipitation estimation from fengyun-4A satellite imagery. *Remote Sens.* 14 (12), 2925.
- Goodwin, N.R., Coops, N.C., Wulder, M.A., Gillanders, S., Schroeder, T.A., Nelson, T., 2008. Estimation of insect infestation dynamics using a temporal sequence of Landsat data. *Remote Sens. Environ.* 112 (9), 3680–3689.
- Hamilton, D., Brothers, K., McCall, C., Gautier, B., Shea, T., 2021. Mapping forest burn extent from hyperspatial imagery using machine learning. *Remote Sens.* 13 (19), 3843.
- Hislop, S., Jones, S., Soto-Berelov, M., Skidmore, A., Haywood, A., Nguyen, T.H., 2018. Using landsat spectral indices in time-series to assess wildfire disturbance and recovery. *Remote Sens.* 10 (3), 460.
- Hollmann, R., Merchant, C.J., Saunders, R., Downy, C., Buchwitz, M., Cazenave, A., Chuvieco, E., Defourny, P., de Leeuw, G., Forsberg, R., et al., 2013. The ESA climate change initiative: Satellite data records for essential climate variables. *Bull. Am. Meteorol. Soc.* 94 (10), 1541–1552.
- Hoover, K., Hanson, L.A., 2021. *Wildfire Statistics*. Technical Report, Congressional Research Service.
- Hossain, M.S., Betts, J.M., Paplinski, A.P., 2021. Dual focal loss to address class imbalance in semantic segmentation. *Neurocomputing* 462, 69–87.
- Hu, X., Ban, Y., Nascetti, A., 2021. Uni-temporal multispectral imagery for burned area mapping with deep learning. *Remote Sens.* 13 (8), 1509.
- Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 7132–7141.
- Huete, A.R., 1988. A soil-adjusted vegetation index (SAVI). *Remote Sens. Environ.* 25 (3), 295–309.
- Jaderberg, M., Simonyan, K., Zisserman, A., et al., 2015. Spatial transformer networks. In: *Advances in Neural Information Processing Systems*, vol. 28.
- Jadon, S., 2020. A survey of loss functions for semantic segmentation. In: *2020 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology*. CIBCB, IEEE, pp. 1–7.
- Jiang, Z., Huete, A.R., Didan, K., Miura, T., 2008. Development of a two-band enhanced vegetation index without a blue band. *Remote Sens. Environ.* 112 (10), 3833–3845.
- Jones, M.W., Abatzoglou, J.T., Veraverbeke, S., Andela, N., Lasslop, G., Forkel, M., Smith, A.J., Burton, C., Betts, R.A., van der Werf, G.R., et al., 2022. Global and regional trends and drivers of fire under climate change. *Rev. Geophys.* 60 (3), e2020RG000726.
- Katagis, T., Gitas, I., 2022. Assessing the accuracy of MODIS MCD64A1 C6 and FireCCI51 Burned Area products in Mediterranean ecosystems. *Remote Sens.-Basel*, 14, 602.
- Key, C.H., Benson, N.C., 2006. *Landscape Assessment (LA) Sampling and Analysis Methods*. USDA Forest Service Gen. Technical Report, Tech. Rep. RMRS-GTR-164-CD.
- Khryashchev, V., Larionov, R., 2020. Wildfire segmentation on satellite images using deep learning. In: *2020 Moscow Workshop on Electronic and Networking Technologies*. MWENT, IEEE, pp. 1–5.
- Knopp, L., Wieland, M., Rättich, M., Martinis, S., 2020. A deep learning approach for burned area segmentation with Sentinel-2 data. *Remote Sens.* 12 (15), 2422.
- Lasko, K., 2021. Incorporating Sentinel-1 SAR imagery with the MODIS MCD64A1 burned area product to improve burn date estimates and reduce burn date uncertainty in wildland fire mapping. *Geocarto Int.* 36 (3), 340–360.
- Li, X., Sun, X., Meng, Y., Liang, J., Wu, F., Li, J., 2019. Dice loss for data-imbalanced NLP tasks. *arXiv preprint arXiv:1911.02855*.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2017. Focal loss for dense object detection. In: *Proceedings of the IEEE International Conference on Computer Vision*. pp. 2980–2988.

- Lizundia-Loiola, J., Otón, G., Ramo, R., Chuvieco, E., 2020. A spatio-temporal active-fire clustering approach for global burned area mapping at 250 m from MODIS data. *Remote Sens. Environ.* 236, 111493.
- Luft, H., Schillaci, C., Ceccherini, G., Vieira, D., Lipani, A., 2022. Deep learning Based Burnt Area mapping using Sentinel 1 for the Santa Cruz mountains lightning complex (CZU) and creek fires 2020. *Fire* 5 (5), 163.
- Martins, V., Roy, D., Huang, H., Boschetti, L., Zhang, H., Yan, L., 2022. Deep learning high resolution burned area mapping by transfer learning from landsat-8 to PlanetScope. *Remote Sens. Environ.* 280, 113203.
- Maskouni, F.H., Seydi, S.T., 2021. Forest burned area mapping using bi-temporal Sentinel-2 imagery based on a convolutional neural network: Case study in Golestan Forest. *Eng. Proc.* 10 (1), 6.
- Michel, C., Lioussé, C., Grégoire, J.-M., Tansey, K., Carmichael, G., Woo, J.-H., 2005. Biomass burning emission inventory from burnt area data given by the SPOT-vegetation system in the frame of TRACE-P and ACE-Asia campaigns. *J. Geophys. Res.: Atmos.* 110 (D9).
- Monaco, S., Greco, S., Farasin, A., Colomba, L., Apiletti, D., Garza, P., Cerquitelli, T., Baralis, E., 2021. Attention to fires: Multi-channel deep learning models for wildfire severity prediction. *Appl. Sci.* 11 (22), 11060.
- Moritz, M.A., Batllori, E., Bradstock, R.A., Gill, A.M., Handmer, J., Hessburg, P.F., Leonard, J., McCaffrey, S., Odion, D.C., Schoennagel, T., et al., 2014. Learning to coexist with wildfire. *Nature* 515 (7525), 58–66.
- Oh, S.H., Ghyme, S.W., Jung, S.K., Kim, G.-W., 2020. Early wildfire detection using convolutional neural network. In: *International Workshop on Frontiers of Computer Vision*. Springer, pp. 18–30.
- Oktay, O., Schlemper, J., Folgoc, L.L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N.Y., Kainz, B., et al., 2018. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*.
- Park, M., Tran, D.Q., Jung, D., Park, S., 2020. Wildfire-detection method using DenseNet and CycleGAN data augmentation-based remote camera imagery. *Remote Sens.* 12 (22), 3715.
- Petropoulos, G.P., Kontoes, C., Keramitsoglou, I., 2011. Burnt area delineation from a uni-temporal perspective based on landsat TM imagery classification using support vector machines. *Int. J. Appl. Earth Obs. Geoinf.* 13 (1), 70–80.
- Pinto, M.M., Libonati, R., Trigo, R.M., Trigo, I.F., DaCamara, C.C., 2020. A deep learning approach for mapping and dating burned areas using temporal sequences of satellite images. *ISPRS J. Photogramm. Remote Sens.* 160, 260–274.
- Sakr, G.E., Elhajj, I.H., Mitri, G., Wejinya, U.C., 2010. Artificial intelligence for forest fire prediction. In: *2010 IEEE/ASME International Conference on Advanced Intelligent Mechatronics*. IEEE, pp. 1311–1316.
- Seydi, S.T., Hasanlou, M., Chanussot, J., 2021. DSMNN-Net: A deep siamese morphological neural network model for burned area mapping using multispectral Sentinel-2 and hyperspectral PRISMA images. *Remote Sens.* 13 (24), 5138.
- Seydi, S.T., Hasanlou, M., Chanussot, J., 2022a. Burnt-net: Wildfire burned area mapping with single post-fire Sentinel-2 data and deep learning morphological neural network. *Ecol. Indic.* 140, 108999.
- Seydi, S.T., Hasanlou, M., Chanussot, J., 2022b. A quadratic morphological deep neural network fusing radar and optical data for the mapping of burned areas. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 15, 4194–4216.
- Seydi, S.T., Sadegh, M., 2023. Improved burned area mapping using monotemporal Landsat-9 imagery and convolutional shift-transformer. *Measurement* 216, 112961.
- Shirvani, Z., Abdi, O., Goodman, R.C., 2023. High-resolution semantic segmentation of woodland fires using residual attention UNet and time series of Sentinel-2. *Remote Sens.* 15 (5), 1342.
- Storey, E.A., Stow, D.A., O'Leary, J.F., 2016. Assessing postfire recovery of chamise chaparral using multi-temporal spectral vegetation index trajectories derived from Landsat imagery. *Remote Sens. Environ.* 183, 53–64.
- Stroppiana, D., Bordogna, G., Boschetti, M., Carrara, P., Boschetti, L., Brivio, P.A., 2011. Positive and negative information for assessing and revising scores of burn evidence. *IEEE Geosci. Remote Sens. Lett.* 9 (3), 363–367.
- Verbesselt, J., Hyndman, R., Newnham, G., Culvenor, D., 2010. Detecting trend and seasonal changes in satellite image time series. *Remote Sens. Environ.* 114 (1), 106–115.
- Verstraete, M., Pinty, B., 1996. Designing optimal spectral indexes for remote sensing applications. *IEEE Trans. Geosci. Remote Sens.* 34 (5), 1254–1265.
- Wang, Q., Adiku, S., Tenhunen, J., Granier, A., 2005. On the relationship of NDVI with leaf area index in a deciduous forest site. *Remote Sens. Environ.* 94 (2), 244–255.
- Wang, L., Wang, C., Sun, Z., Chen, S., 2020. An improved dice loss for pneumothorax segmentation by mining the information of negative areas. *IEEE Access* 8, 167939–167949.
- Yeung, M., Sala, E., Schönlieb, C.-B., Rundo, L., 2022. Unified focal loss: Generalising dice and cross entropy-based losses to handle class imbalanced medical image segmentation. *Comput. Med. Imaging Graph.* 95, 102026.
- Zhang, P., Ban, Y., Nascetti, A., 2023. Total-variation regularized U-Net for wildfire burned area mapping based on Sentinel-1 C-band SAR backscattering data. *ISPRS J. Photogramm. Remote Sens.* 203, 301–313.
- Zhang, P., Hu, X., Ban, Y., 2022. Wildfire-S1S2-Canada: A large-scale Sentinel-1/2 Wildfire Burned Area mapping dataset based on the 2017–2019 wildfires in Canada. In: *IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, pp. 7954–7957.
- Zhao, R., Qian, B., Zhang, X., Li, Y., Wei, R., Liu, Y., Pan, Y., 2020. Rethinking dice loss for medical image segmentation. In: *2020 IEEE International Conference on Data Mining. ICDM, IEEE*, pp. 851–860.
- Zheng, X., Chen, T., 2021. High spatial resolution remote sensing image segmentation based on the multiclassification model and the binary classification model. *Neural Comput. Appl.* 1–8.